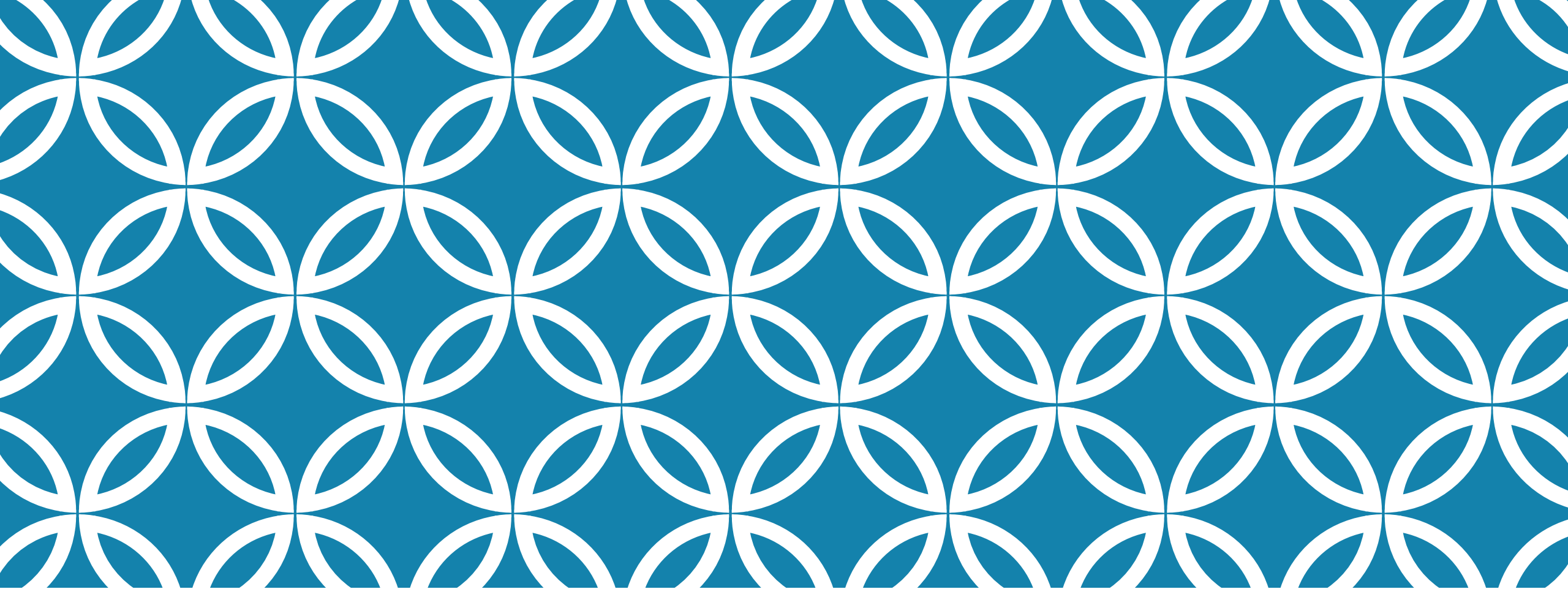




INTRODUCTION TO SWITCHING

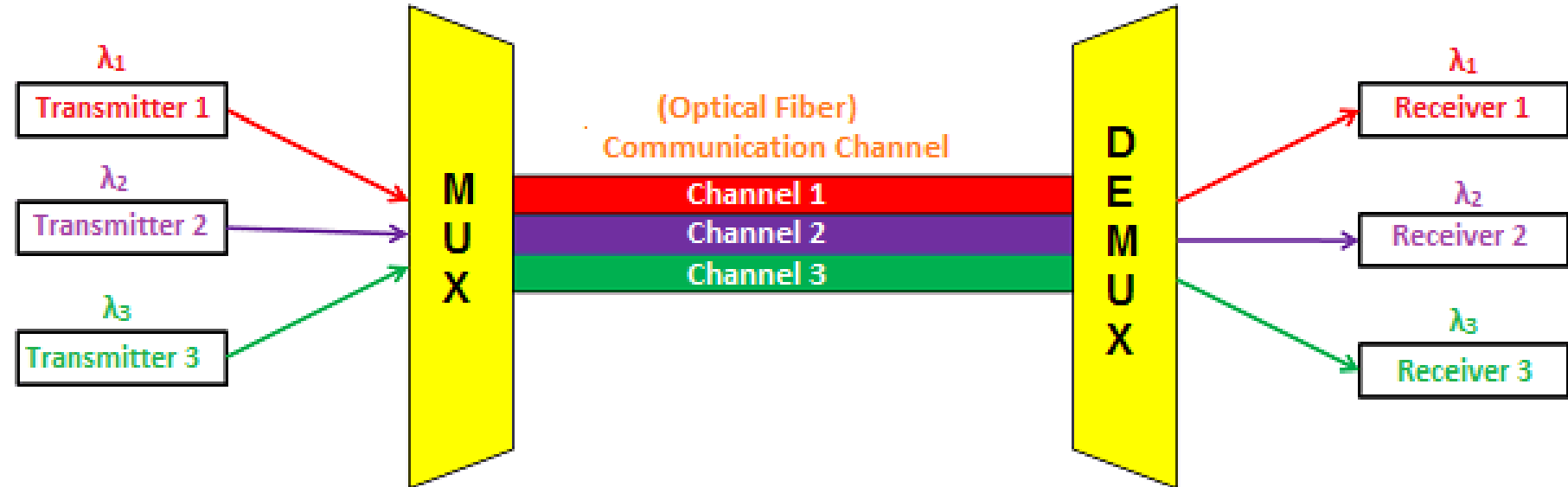
SWITCHING NETWORKS

- Long distance transmission is typically done over a **network of switched nodes**
- Nodes not concerned with content of data.
- **End devices** are stations (*Computer, Terminal, Phone, etc.*)
- A collection of **nodes** and **connections** is a **communications network**.
- Data routed by being **switched** from node to node.

NODES

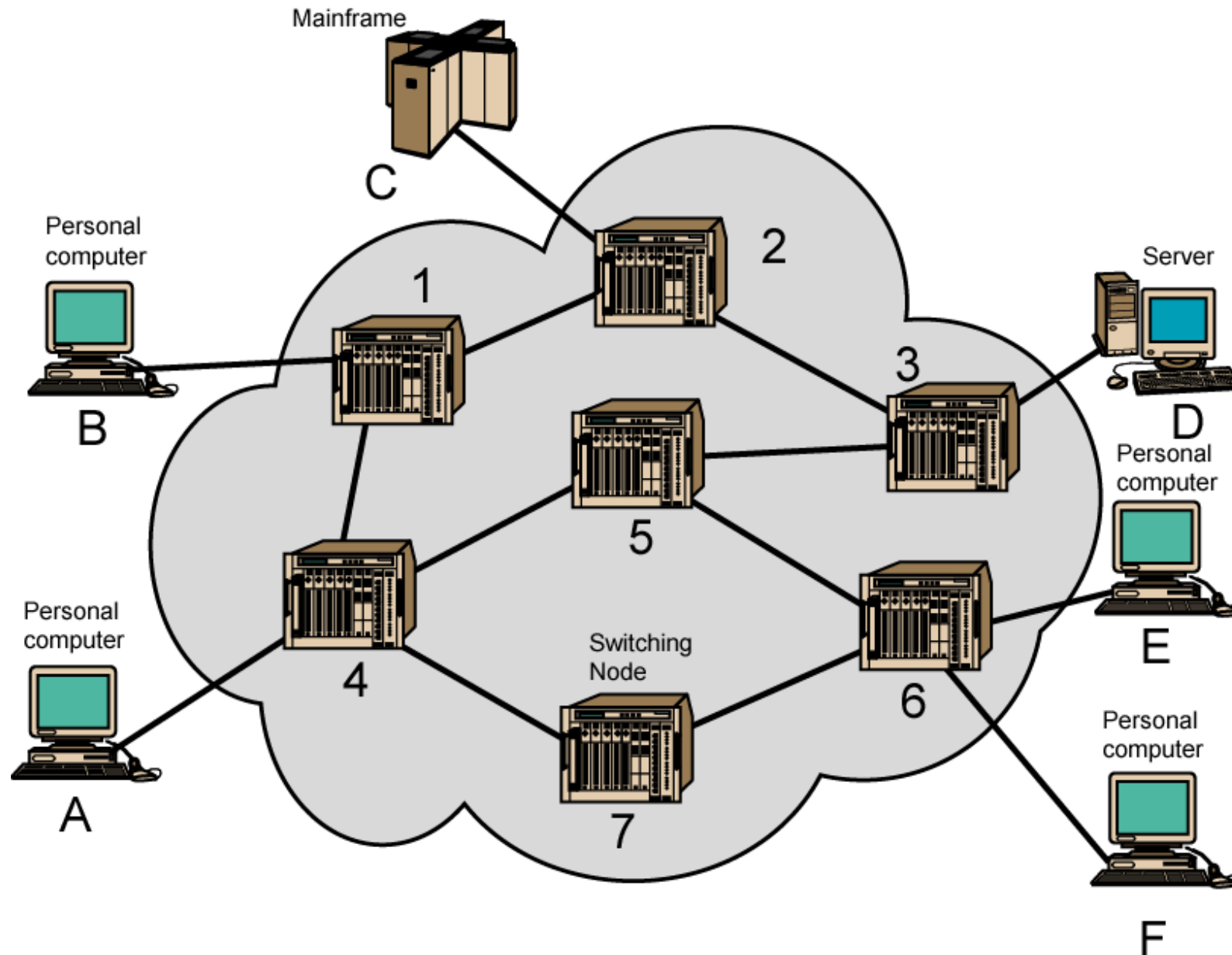
- Nodes may connect to other nodes only, or to stations and other nodes

Node to Node links usually **multiplexed**.



- Two different switching technologies:
 - Circuit Switching
 - Packet Switching

SIMPLE SWITCHED NETWORK



CIRCUIT SWITCHING

- Dedicated communication path between two stations

Three phases

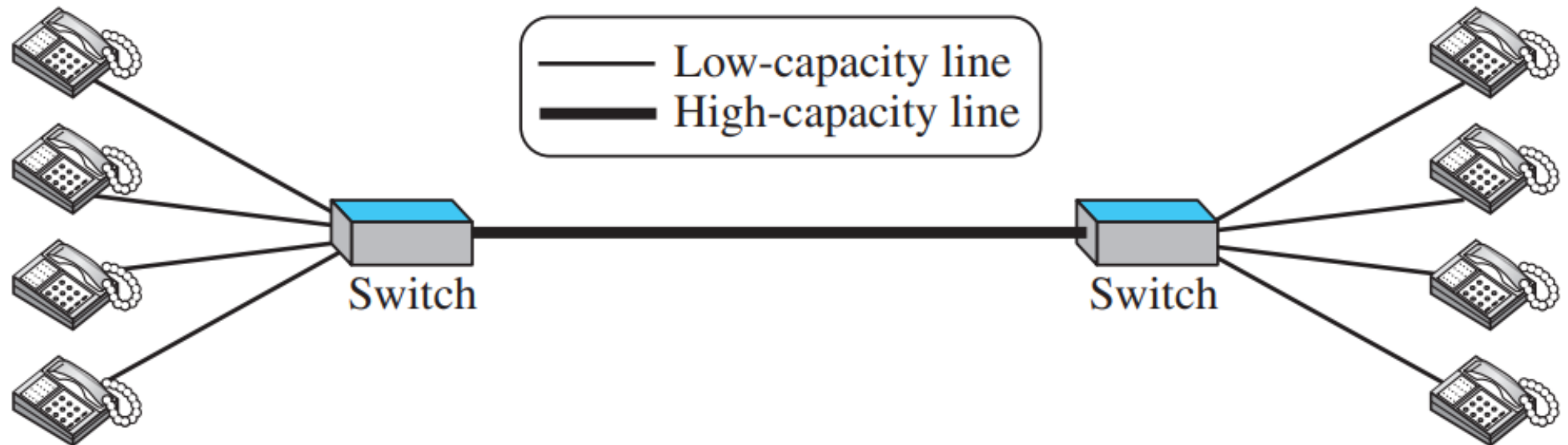
- **Circuit Establishment**

- **Data Transfer**

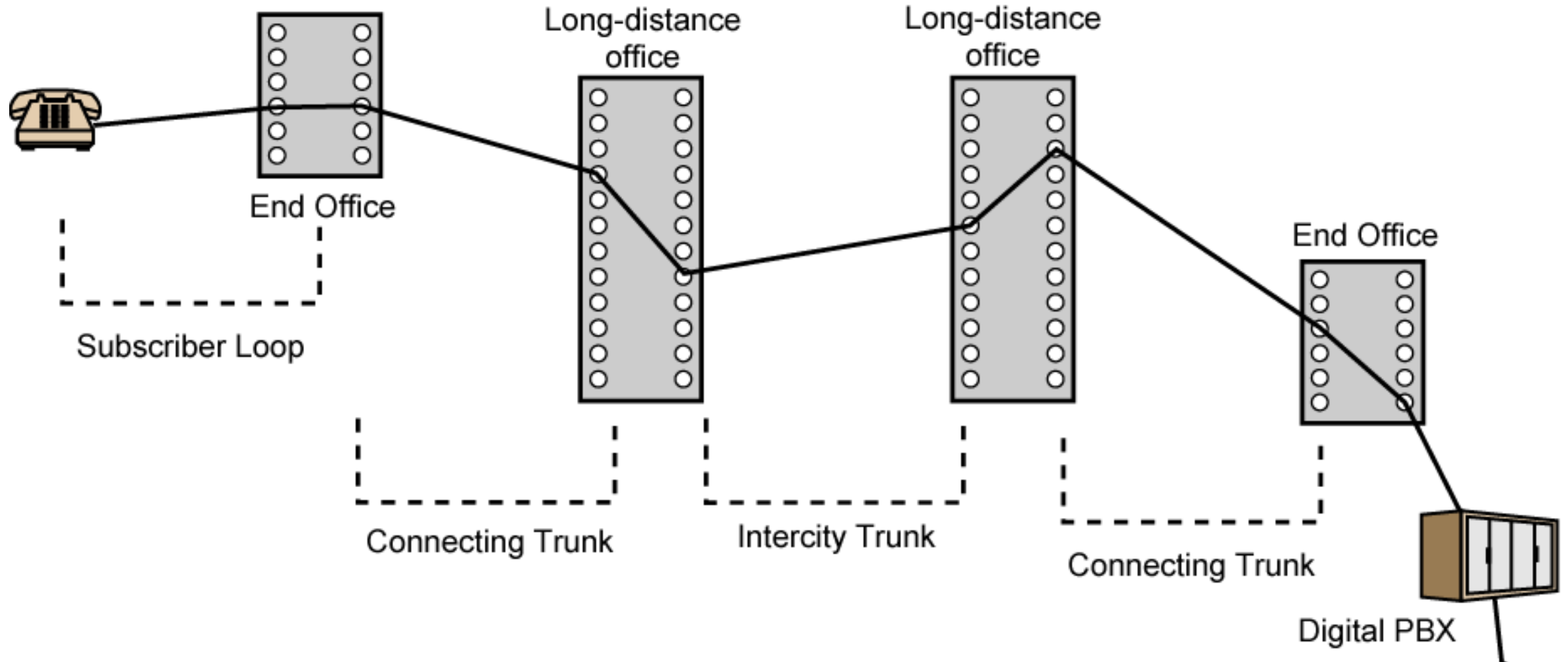
- **Circuit Disconnect**

- Must have **switching capacity** and **channel capacity** to establish the connection
- Must have intelligence to work out **routing**.
- Circuit switching takes place at the **physical layer**.

- It is inefficient
 - Channel capacity dedicated for the duration of the connection
 - If there is no data, capacity is wasted
- Set up (connection) takes time
- Once connected, the transfer is transparent
- Developed for **voice traffic (phone)**



PUBLIC CIRCUIT SWITCHED NETWORK



Private Branch Exchange (PBX)

TELECOMMUNICATION COMPONENTS

Subscriber - Devices attached to the network

Subscriber line – Link between the **subscriber** and the **network**

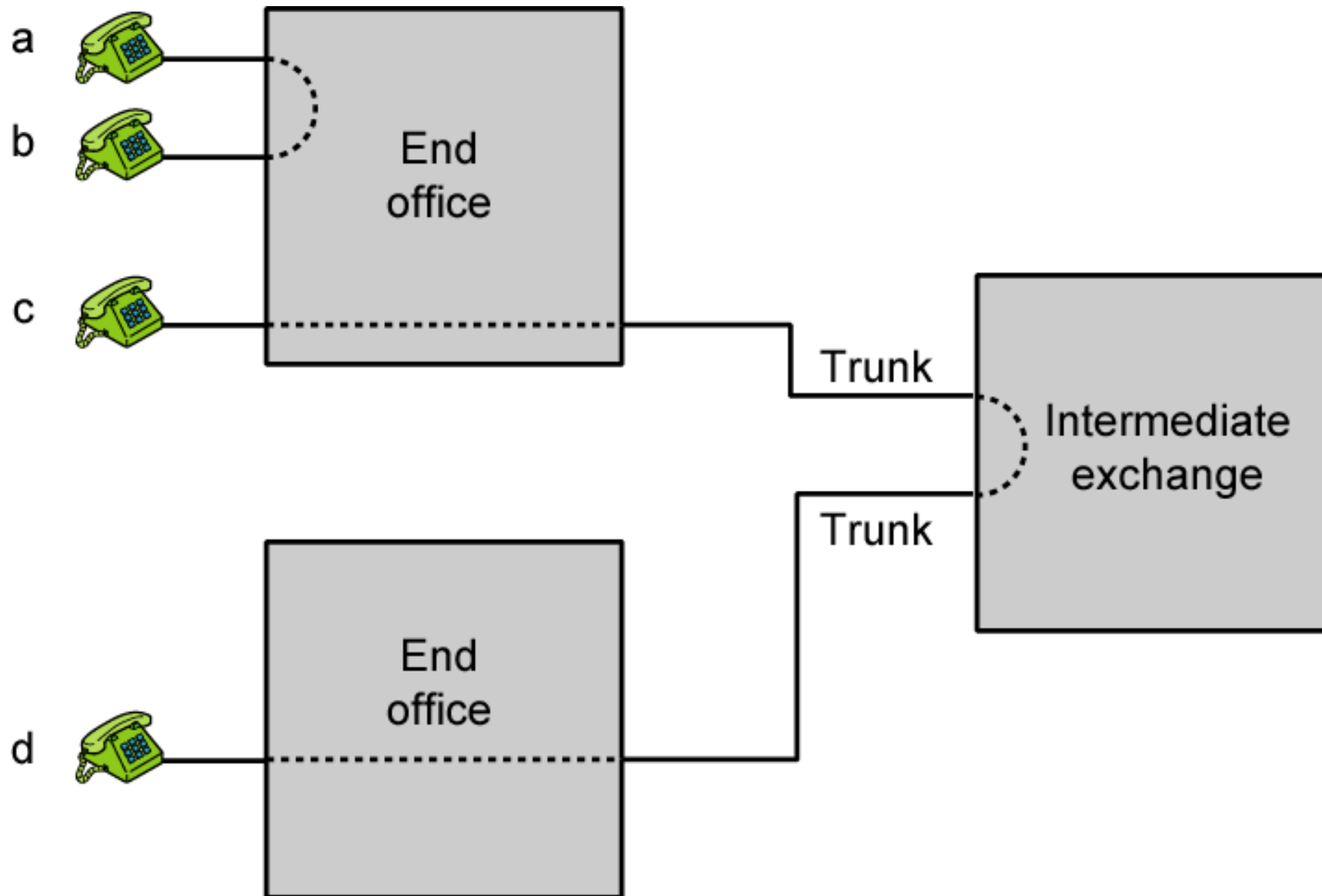
- Local Loop (or) Subscriber loop
- Uses Twisted-Pair wire
- Few km up to a few tens of km

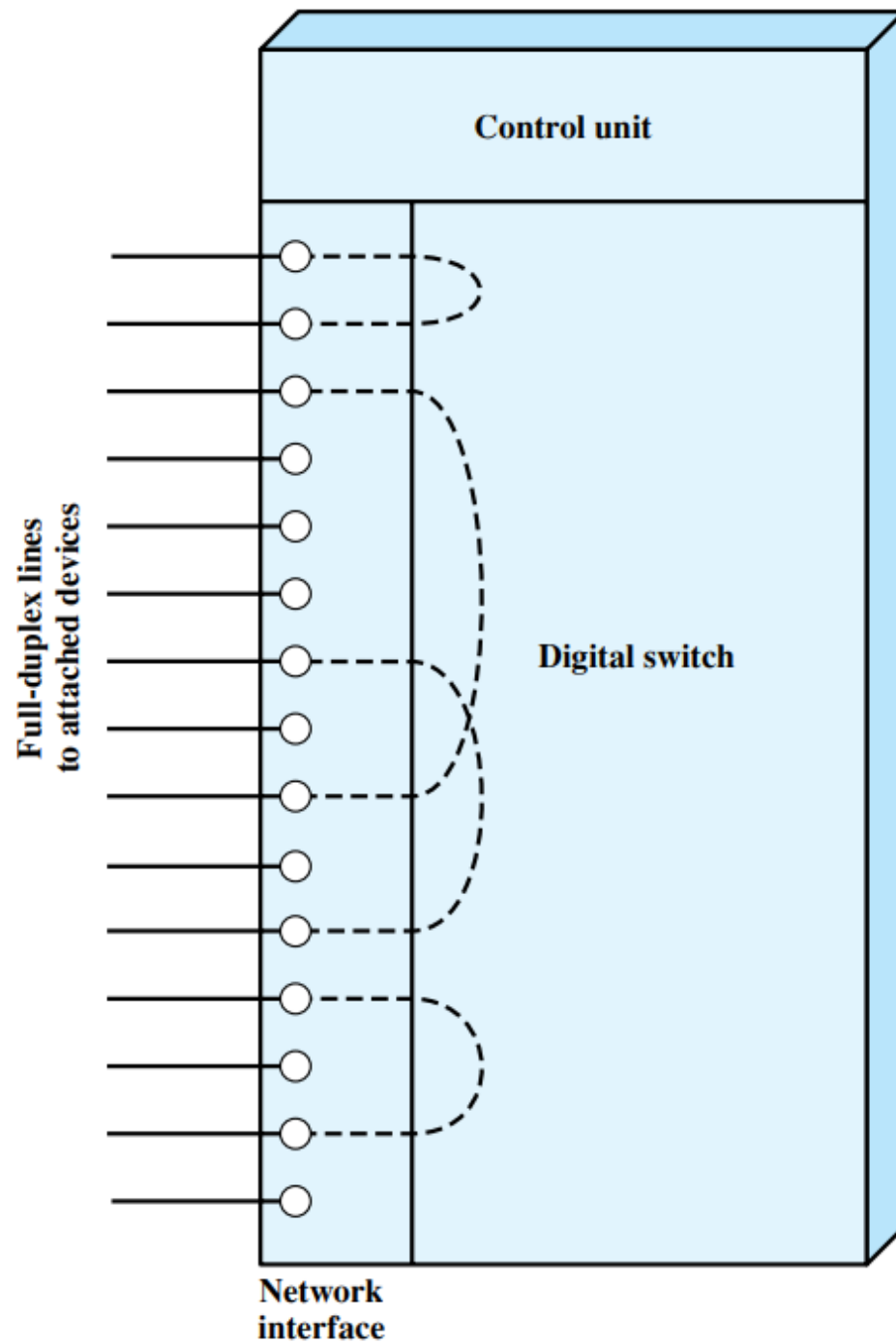
Exchange - Switching centers in the network

End office - supports subscribers

Trunks - Branches between exchanges (Multiplexed)

CIRCUIT ESTABLISHMENT





ELEMENTS OF A CIRCUIT-SWITCH NODE

CIRCUIT SWITCHING CONCEPTS

- **Digital Switch**

- Used to provide a transparent signal path between devices

- **Network Interface**

- It holds functions and hardware needed to connect digital devices

- **Control Unit**

- ***Establishes connections***
 - Generally, on demand
 - Handle and acknowledge requests
 - Determine if the destination is free
 - Construct path
 - ***Maintain connection***
 - ***Disconnect***

BLOCKING OR NON-BLOCKING

Blocking

- A network is unable to connect stations because all paths are in use. It is allowed in a blocking network.
- Used on voice systems
 - Short-duration calls

Non-blocking

- Permits all stations to connect (in pairs) at once
- Used for some data connections

- **Circuit Switching** is designed for voice calls
 - Resources dedicated to a particular call
 - Much of the time a data connection is idle
 - Data rate is fixed
 - Both ends must operate at the same rate

PACKET SWITCHING



Data transmitted in small packets

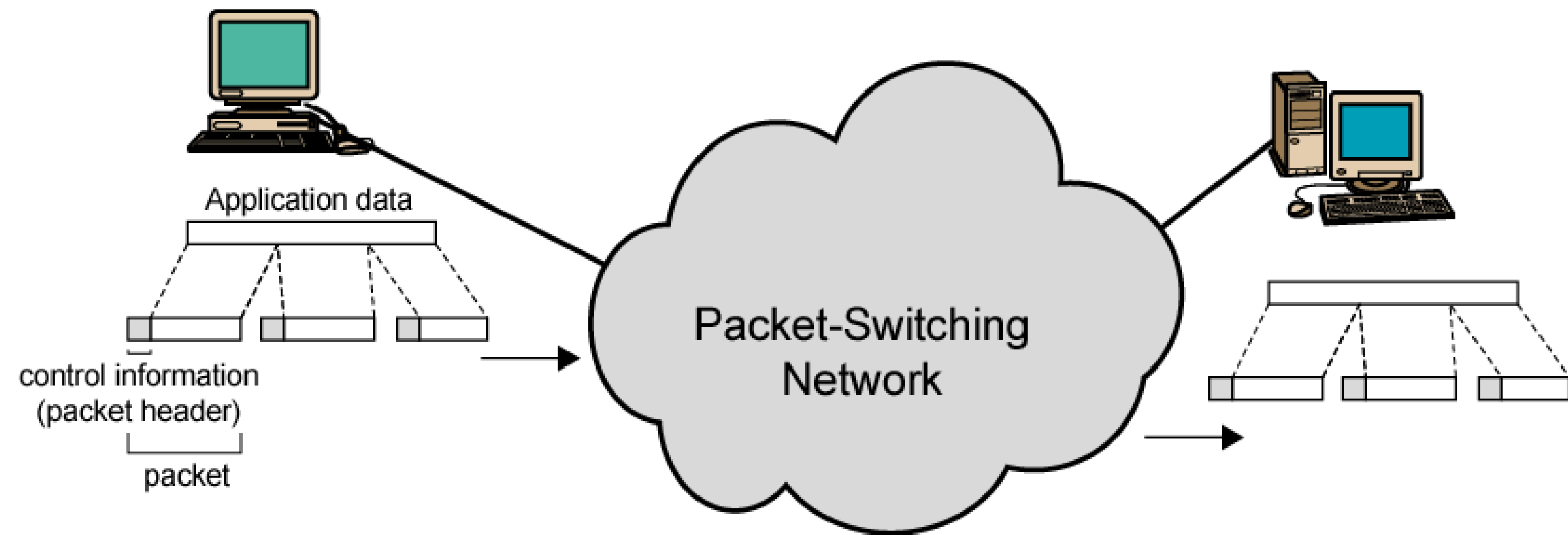
- Typically, 1000 octets *octet – 8 bits*
- Longer messages split into series of packets
- Each packet contains a **portion of user data** plus some **control information**

Control information

- **Routing (addressing)** information

Packets are received, stored briefly (buffered), and passed on to the next node

- Store and Forward



ADVANTAGES

➤ **Line efficiency**

- Single node to node link can be shared by many packets over time
- Packets queued and transmitted as fast as possible

➤ **Data rate conversion**

- Each station connects to the local node at its own speed
- Nodes buffer data if required to equalize rates

➤ **Packets are accepted even when network is busy**

- Delivery may slow down

➤ **Priorities can be used**

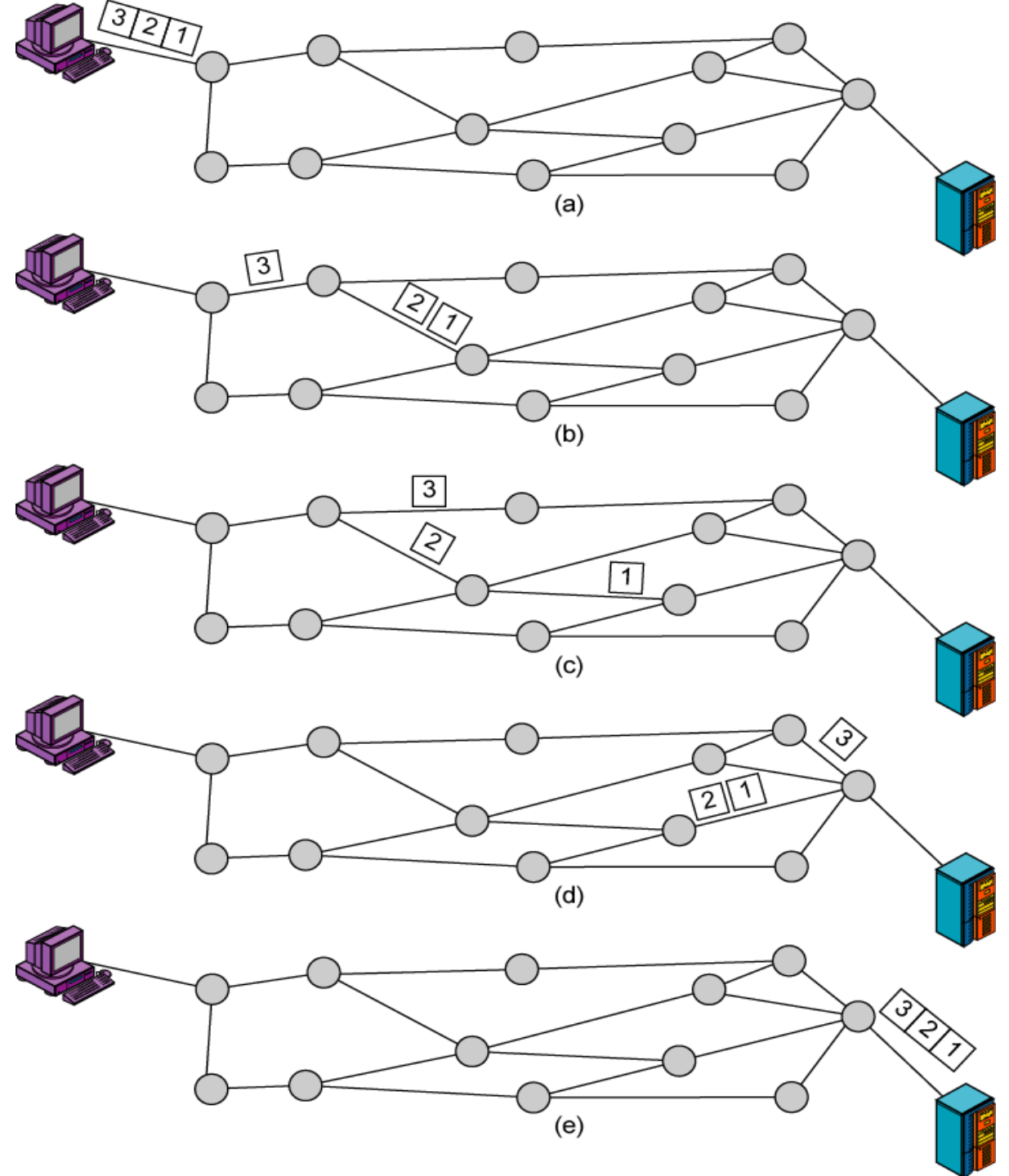
SWITCHING TECHNIQUE

- Station breaks long message into packets
- Packets sent one at a time to the network
- Packets handled in two ways
 - **Datagram**
 - **Virtual circuit**

DATAGRAM

- Each packet treated independently
- Packets can take any practical route
- Packets may arrive out of order
- Packets may go missing
- Up to receiver to re-order packets and recover from missing packets

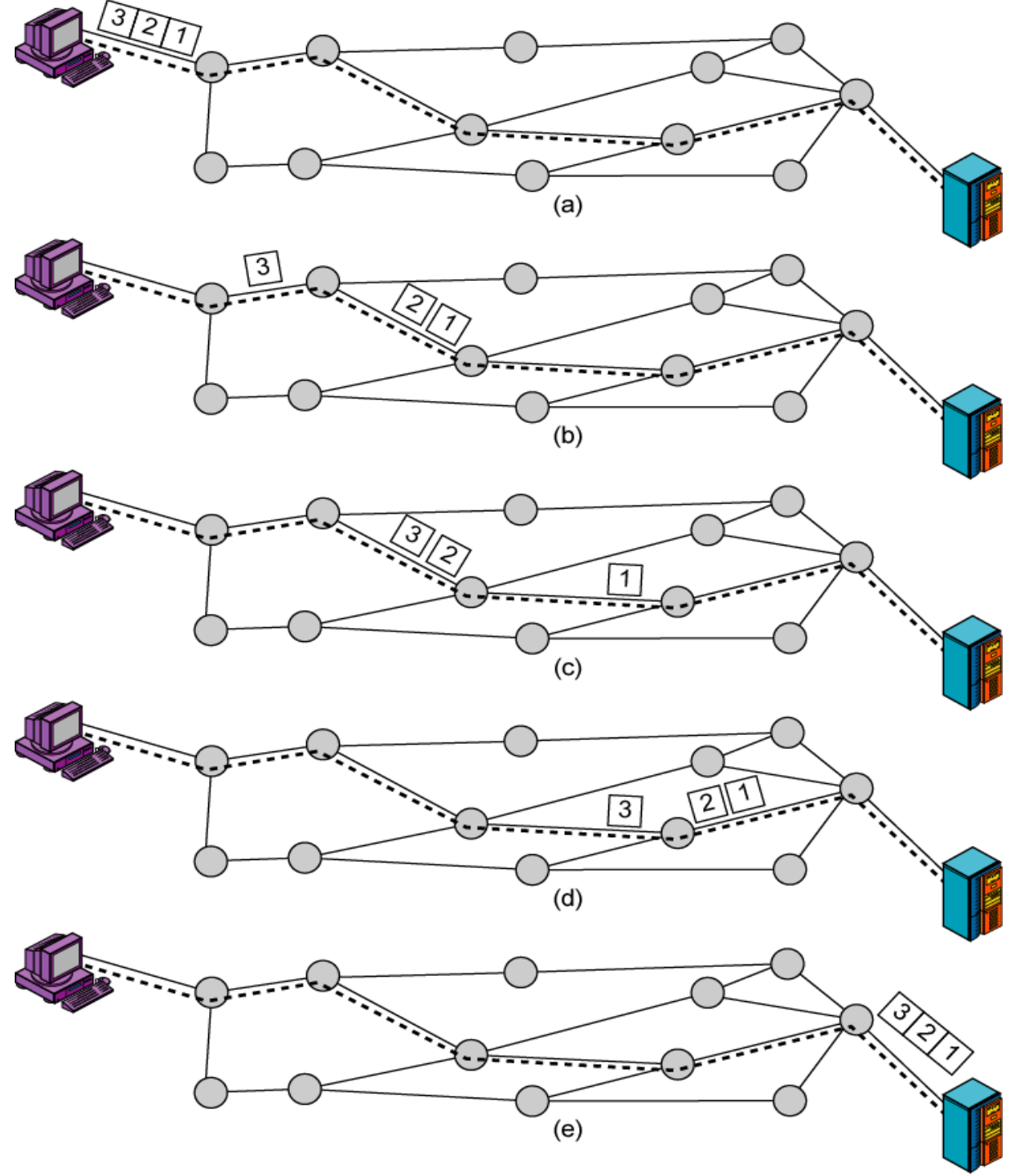
DATAGRAM DIAGRAM



VIRTUAL CIRCUIT

- Preplanned route established before any packets are sent
- **Call request** and **call accept** packets establish the connection (handshake)
- Each packet contains a **virtual circuit identifier** instead of the destination address
- No routing decisions are required for each packet
- Clear request to drop circuit
- Not a dedicated path

VIRTUAL CIRCUIT DIAGRAM



VIRTUAL CIRCUITS V DATAGRAM

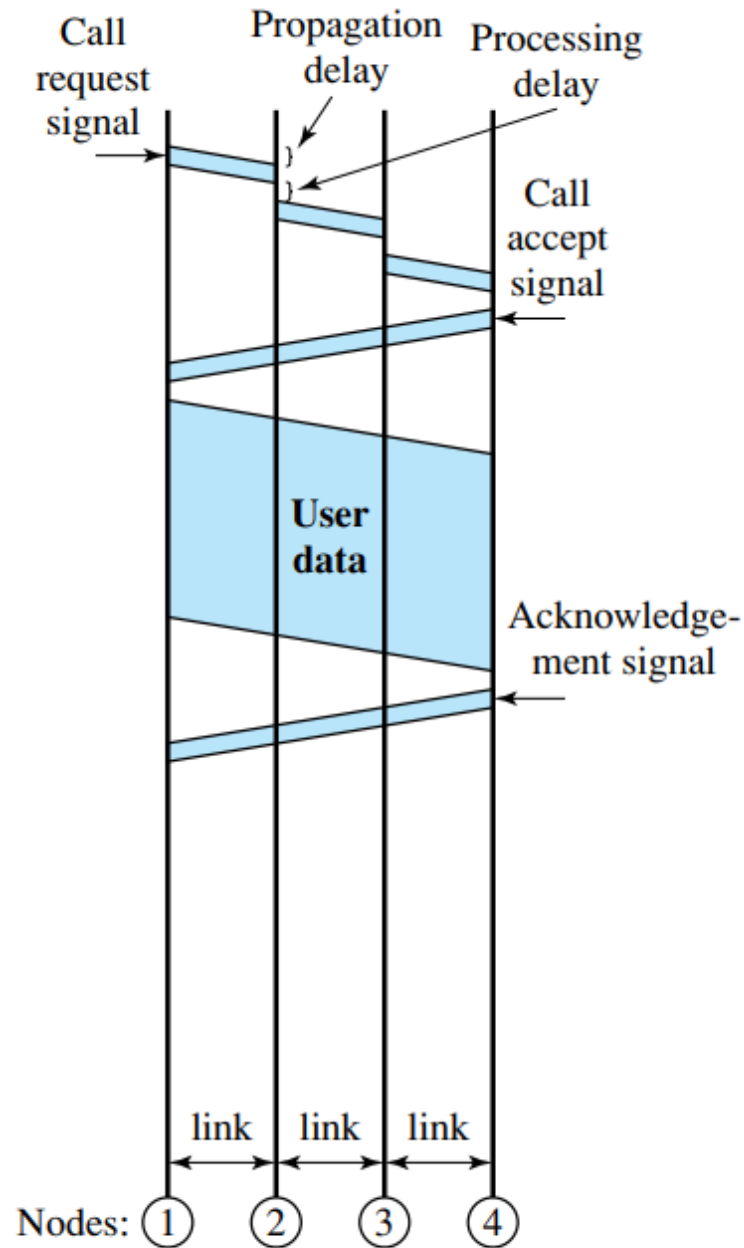
Virtual circuits

- Network can provide **sequencing** and **error control**
- Packets are forwarded more quickly
 - No routing decisions to make
- Less reliable
 - Loss of a node loses all circuits through that node

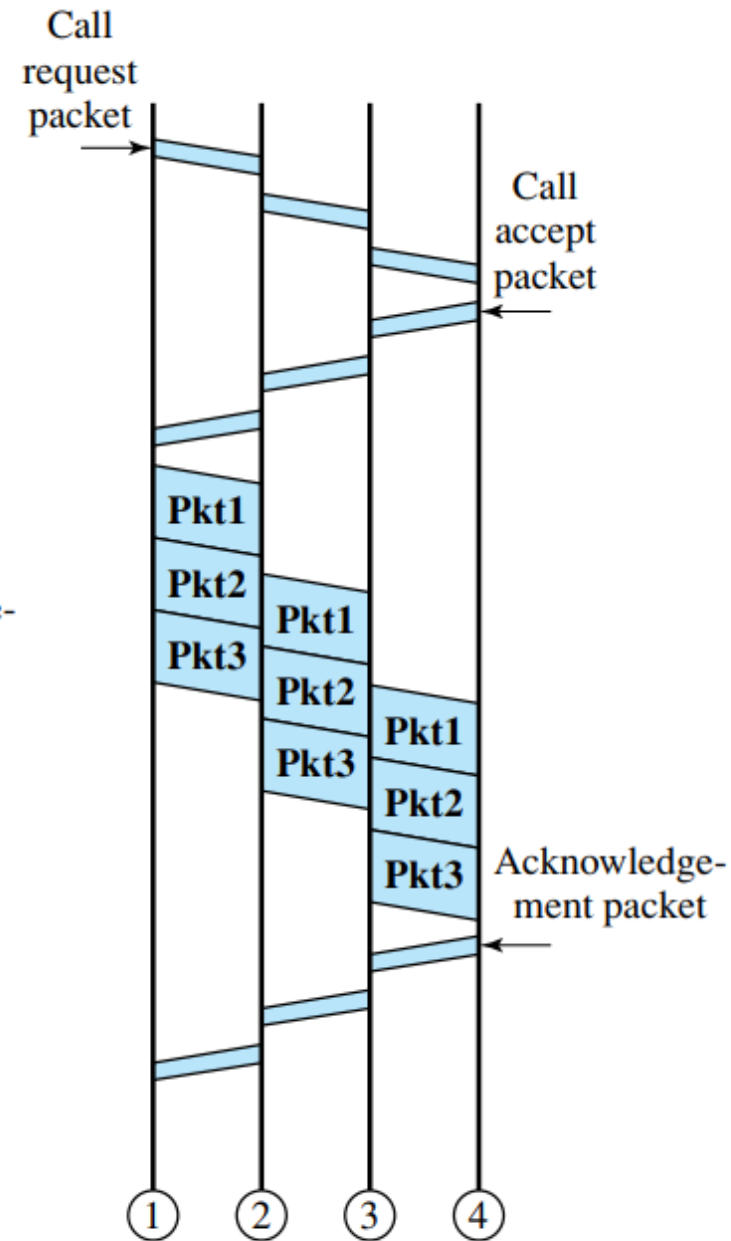
Datagram

- No call setup phase, better if few packets
- More flexible
 - Routing can be used to avoid congested parts of the network

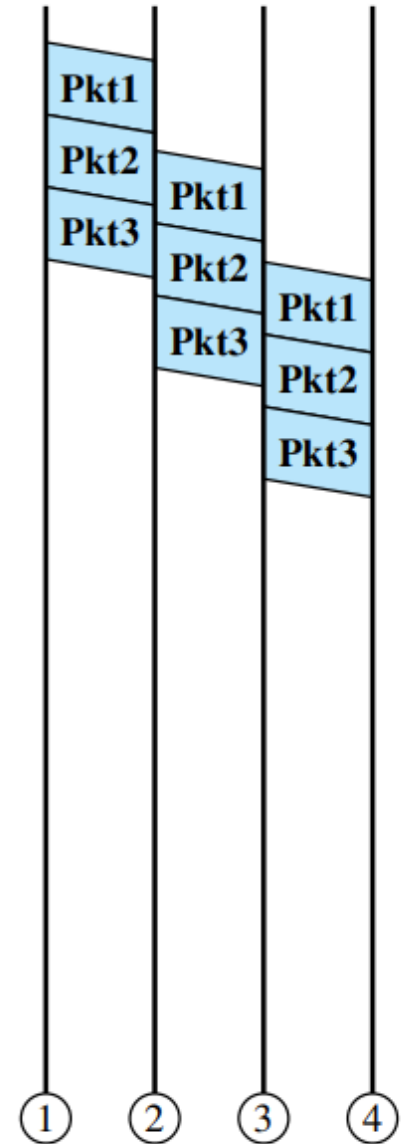
EVENT TIMING



(a) Circuit switching

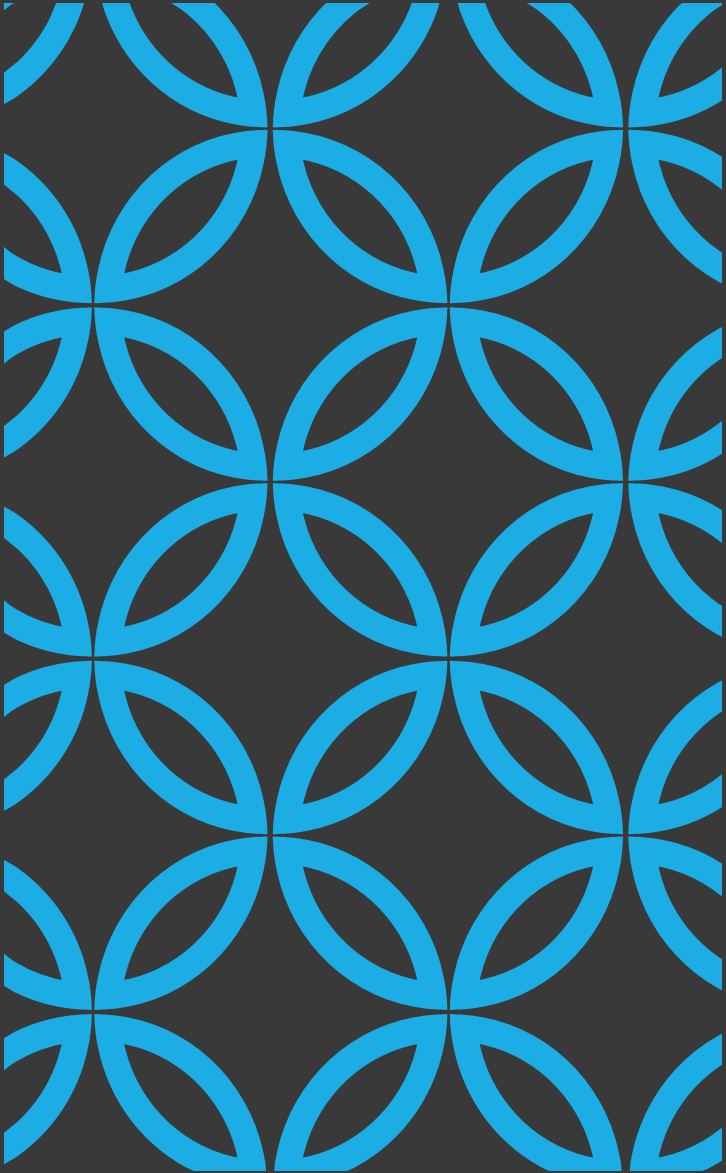


(b) Virtual circuit packet switching



(c) Datagram packet switching

Circuit Switching	Datagram Packet Switching	Virtual Circuit Packet Switching
Dedicated transmission path	No dedicated path	No dedicated path
Continuous transmission of data	Transmission of packets	Transmission of packets
Fast enough for interactive	Fast enough for interactive	Fast enough for interactive
Messages are not stored	Packets may be stored until delivered	Packets stored until delivered
The path is established for entire conversation	Route established for each packet	Route established for entire conversation
Call setup delay; negligible transmission delay	Packet transmission delay	Call setup delay; packet transmission delay
Busy signal if called party busy	Sender may be notified if packet not delivered	Sender notified of connection denial
Overload may block call setup; no delay for established calls	Overload increases packet delay	Overload may block call setup; increases packet delay
Electromechanical or computerized switching nodes	Small switching nodes	Small switching nodes
User responsible for message loss protection	Network may be responsible for individual packets	Network may be responsible for packet sequences
Usually no speed or code conversion	Speed and code conversion	Speed and code conversion
Fixed bandwidth	Dynamic use of bandwidth	Dynamic use of bandwidth
No overhead bits after call setup	Overhead bits in each packet	Overhead bits in each packet

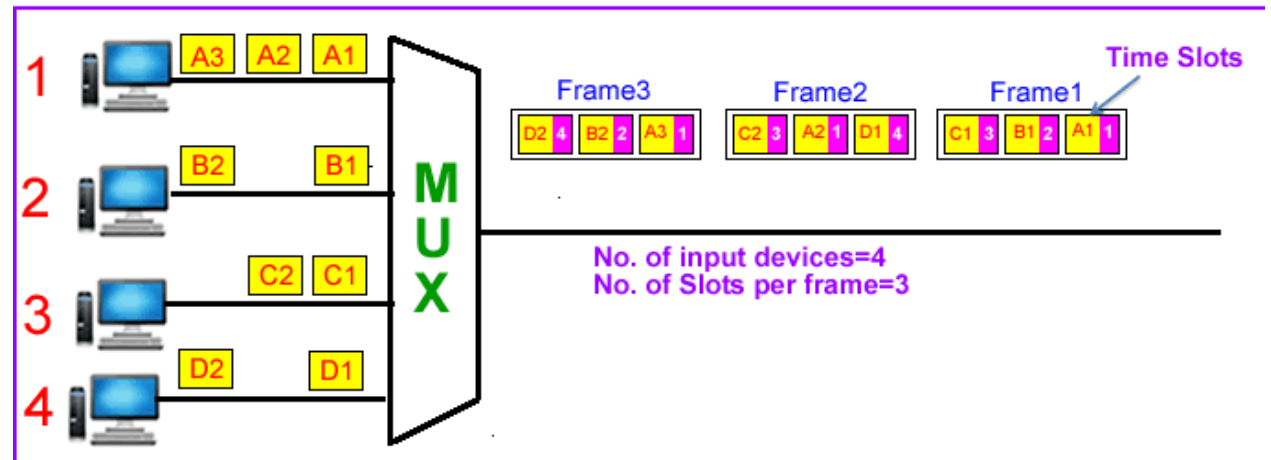
A decorative pattern on the left side of the slide, consisting of a grid of overlapping circles in a light blue color, creating a tessellated effect.

FRAME RELAY

- **Frame Relay** is a high-performance **WAN protocol** that operates at the **physical and data link layers** of the OSI reference model.
- It was originally designed for use across **Integrated Services Digital Network (ISDN)** interfaces.
- It is based on **packet-switched** technology.
- Techniques used in packet-switching technology:

➤ **Variable-length packets**

➤ **Statistical multiplexing**



Devices attached to a Frame Relay WAN fall into the following two general categories:

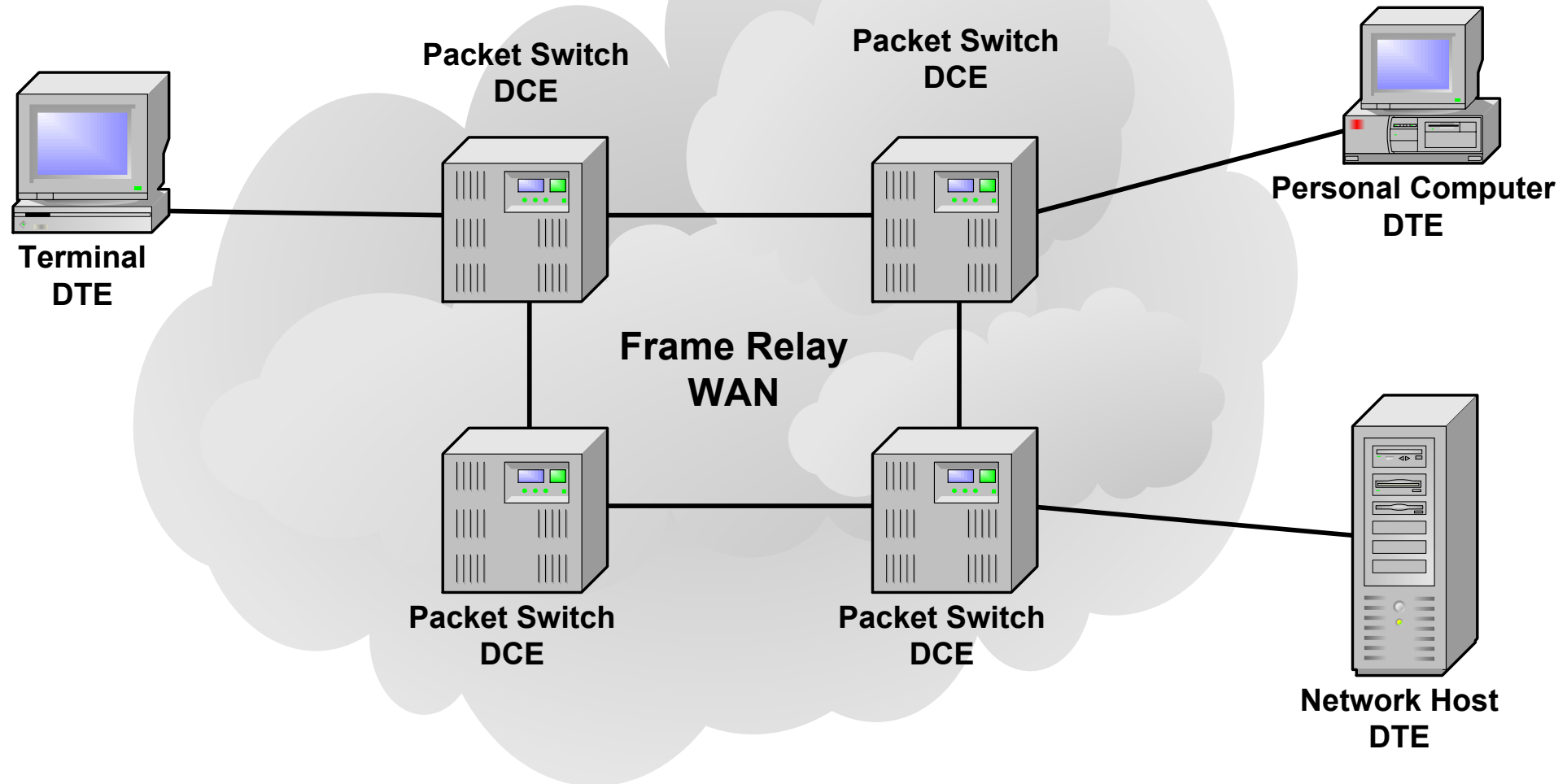
- **Data Terminal Equipment (DTE)**

- Terminals, Personal Computers, Routers, and Bridges typically at the customer location.

- **Data Circuit-terminating Equipment (DCE)**

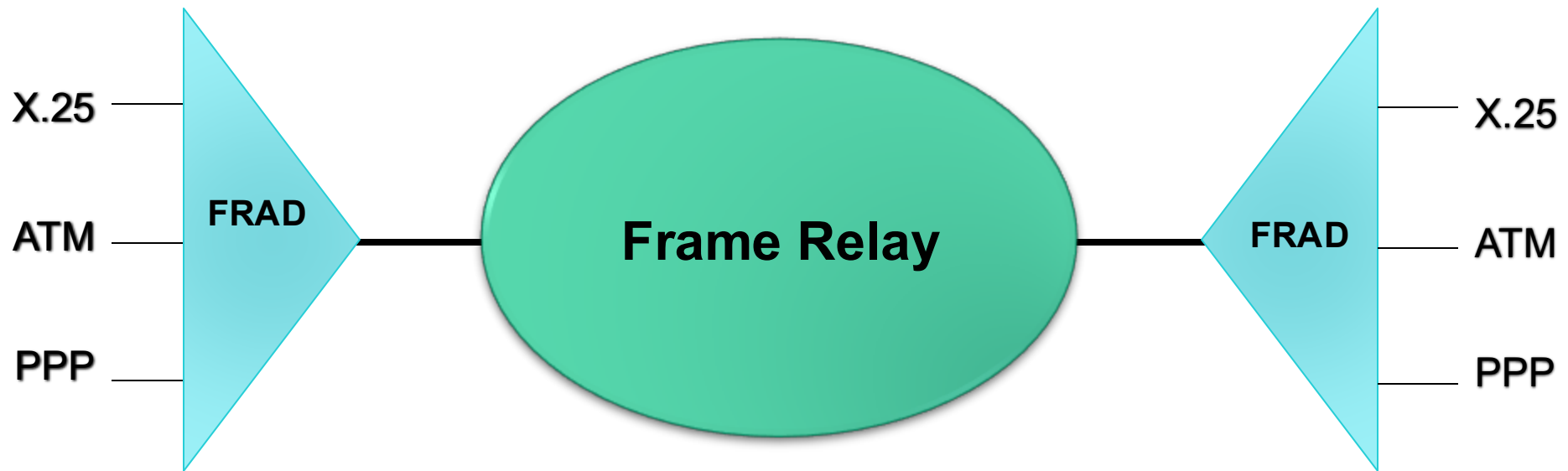
- Typically packet switches owned by the carrier that transmit data through the WAN.

SAMPLE FRAME RELAY NETWORK



FRAME RELAY ASSEMBLER/DISASSEMBLER (FRAD)

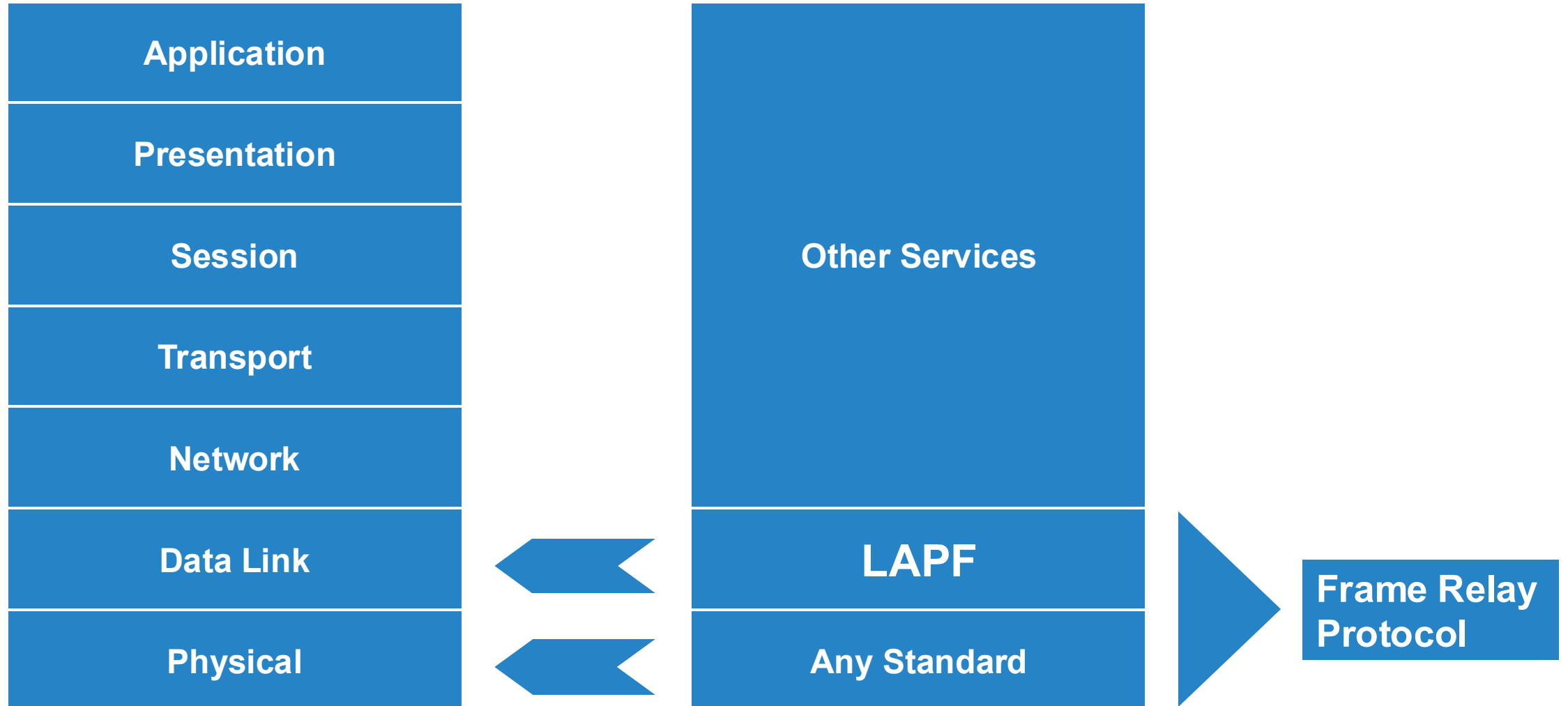
- To handle frames from other protocols a **FRAD** is used to provide conversion to Frame Relay packets.
- A **FRAD** can either be a separate device or part of a router/switch.



Point-to-Point Protocol (PPP)

Asynchronous Transfer Mode (ATM)

FRAME RELAY MAPPING TO OSI MODEL



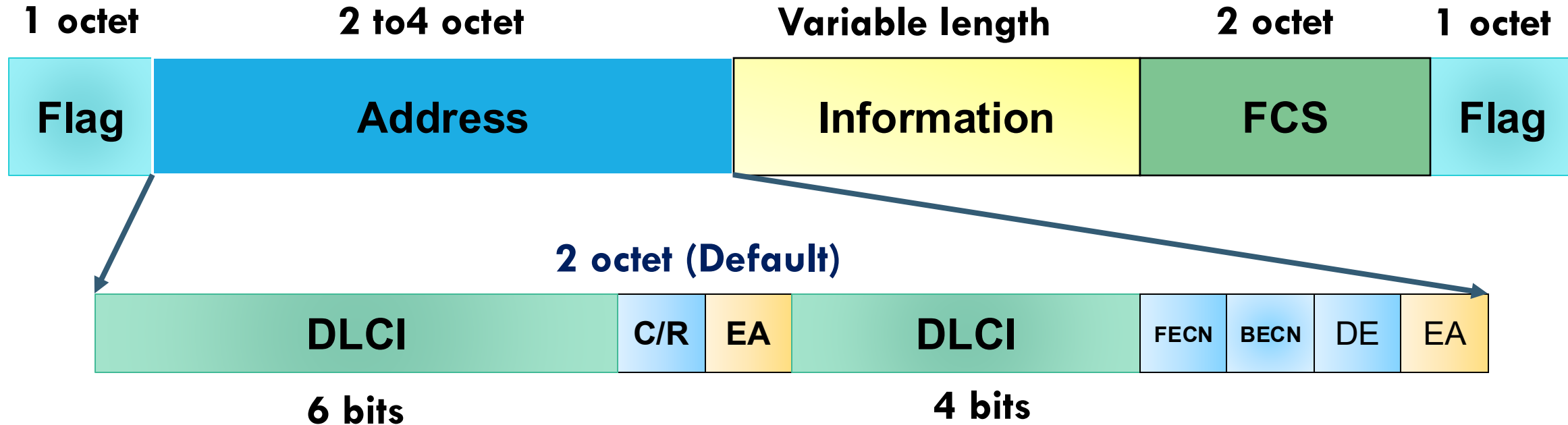
FRAME RELAY PHYSICAL LAYER

- No specific protocol is defined
- Any protocol recognized by ANSI (*American National Standards Institute*) can be implemented

FRAME RELAY DATA LINK LAYER

- **Link Access Protocol for Frame Modes Services (LAPF)** is the protocol defined for Frame Relay Layer 2 services.
- LAPF is a version of HDLC(*High-Level Data Link Control*)
 - Does not provide **flow or error control**
 - Uses Address field for **DLCI (addressing)** as well as for **congestion control**

LAPF FRAME FORMAT



DLCI (10 bits) **Data Link Connection Identifier** is used to identify the **Virtual Circuit number**

C/R (1 bit) Provided for upper layers to determine **commands** and **responses**

EA (1 bit) Determines if this byte is **last byte of address** (0=more, 1=last)

- **FECN (1 bit) Forward Explicit Congestion Notification**
indicates congestion in the direction the frame is traveling
- **BECN (1 bit) Backward Explicit Congestion Notification**
indicates congestion in the opposite direction the frame is traveling
- **DE (1 bit) Discard Eligibility** indicates that a frame is low priority when set

EXTENDED ADDRESSES

- To increase the number of virtual circuits, the **DLCI** can be expanded from **10 bits to 16 bits and 23 bits**.
- The **EA** field is set to **0** to indicate that additional address bytes are present. The last address byte will have a **1** in the **EA** field.

THREE ADDRESS FORMATS

**Two-byte Address
(10-bit DLCI)**

DLCI			C/R	0
DLCI	FECN	BECN	DE	1

**Three-byte Address
(16 bit DLCI)**

DLCI			C/R	0
DLCI	FECN	BECN	DE	0
DLCI			0	1

**Four-byte Address
(23 bit DLCI)**

DLCI			C/R	0
DLCI	FECN	BECN	DE	0
DLCI			0	
DLCI			0	1